

Lead Scoring Case Study – 500 Word Summary Report

This Lead Scoring Case Study was undertaken with the objective of building a predictive model that could identify prospective customers who are most likely to convert into paying customers. The primary business goal was to help the sales team prioritize leads effectively and improve overall conversion rates by focusing on high-quality leads. The project began with understanding the dataset and identifying the target variable, 'Converted'. The initial phase involved extensive data cleaning and preprocessing. Missing values were examined carefully, and columns with a large proportion of missing data or low business relevance were removed. Categorical variables were analyzed and transformed into a suitable format using dummy variable creation so that they could be used by machine learning algorithms. Numerical variables were checked for consistency and scaled where required. After cleaning the data, exploratory data analysis (EDA) was performed to understand the relationships between different variables and lead conversion. Various patterns related to lead source, lead origin, website activity, tags, and customer engagement were examined. This step helped in identifying important features that could influence the conversion probability of a lead. The dataset was then divided into training and testing sets using train-test split. Feature scaling was applied to numerical variables to ensure that all variables were on a comparable scale. A Logistic Regression model was selected because it is interpretable, widely used for binary classification problems, and provides meaningful insights through model coefficients. The model was trained on the training dataset and evaluated using metrics such as accuracy, confusion matrix, classification report, and ROC-AUC score. Predicted probabilities were used to generate lead scores, allowing each lead to be ranked according to its likelihood of conversion. The model coefficients were also analyzed to identify the variables that had the greatest influence on conversion probability. Variables such as lead tags and engagement indicators emerged as strong predictors of successful conversion. One of the most important lessons learned from this assignment was the significance of data preprocessing. The quality of the model depends heavily on how well the data is cleaned and prepared. Another key learning was the importance of balancing business objectives with model performance metrics. Depending on the business scenario, different thresholds can be chosen to optimize either recall or precision. This demonstrated how machine learning models can be aligned with changing business requirements. Overall, this project provided practical experience in the complete machine learning workflow, including data preparation, exploratory analysis, feature engineering, model building, evaluation, and business interpretation. It also highlighted the importance of translating analytical findings into actionable business recommendations. The project strengthened my understanding of predictive analytics and demonstrated how data-driven decision-making can help organizations improve efficiency, prioritize resources, and achieve better business outcomes.